# Assignment -1

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## Part – I

1. In a traditional file system, data is stored in separate files and there is usually no proper connection between those files. Because of this, the same data may be stored again and again, which increases redundancy. It also provides less security, and if many users try to access or update the data at the same time, conflicts can occur. On the other hand, a DBMS stores data in structured tables that are related to each other. It reduces duplicate data, provides better security, and allows multiple users to work on the data simultaneously without creating major problems.

   Example: In a railway reservation system. If two people try to book the last available seat at the same time, a DBMS can ensure that the seat is allotted to only one person and the records are updated correctly. In a traditional file system, there is a possibility that both users may be shown the seat as available, leading to incorrect bookings and data inconsistency.

2. In an ER diagram, a strong entity can be identified by its own primary key, while a weak entity does not have enough attributes to form a primary key by itself. A weak entity depends on a strong entity for its identification and existence.
   The primary key of a weak entity is formed by combining the primary key **of** the strong entity with the partial key (discriminator) of the weak entity. The relationship that connects a weak entity with its strong entity is called an identifying relationship.

3. Phone number should be modeled as a multivalued attribute because one student can have more than one phone number. A simple attribute

can store only one value, while a multivalued attribute can store multiple values for the same entity. Therefore, using a multivalued attribute is the most suitable choice.

4. A candidate key is the smallest set of attributes that can uniquely identify a record. A super key is any set of attributes that can uniquely identify a record, even if it contains extra attributes.
   For the relation Student(roll_no, email, name, branch), both roll_no and email can uniquely identify a student.
   Candidate Keys:
   • roll_no
   • email
   Super Keys that are not Candidate Keys:
   • (roll_no, name)
   • (email, branch)
   These are super keys because they can identify a student uniquely, but they contain extra attributes, so they are not candidate keys.

5. The constraint that is violated is the referential integrity constraint, also called the foreign key constraint.
   This constraint ensures that a foreign key value in one table must exist as a primary key value in the related table. It protects the database from storing invalid references.
   In this case, if the student_id does not exist in the Student table, the insert operation should be rejected and the new row should not be added to the Enrollment table.

6. When a many-to-many relationship is converted into a relational schema, a separate junction table must be created. The primary key of this table is usually formed by combining the primary keys of the two related tables.
   The relationship cannot be represented as a simple column in one table because one record may be related to many records in the other table and vice versa. A single column cannot store multiple related values efficiently, so a separate table is required.

# Part – II

### Part – A

Entities and Key Attributes :-

Customer(Customer_ID, Name, Phone_No)

Restaurant(Restaurant_ID, Restaurant_Name, Location)

MenuItem(MenuItem_ID, Item_Name, Price)

Order(Order_ID, Order_Date)

Relationships :-

Customer – Order : 1:N

One customer can place many orders, but each order belongs to only one customer.

Restaurant – Order : 1:N

A restaurant can receive many orders, but each order is associated with only one restaurant.

Restaurant – MenuItem : 1:N

A restaurant can offer many menu items, but each menu item belongs to one restaurant.

Order – MenuItem : M:N

An order can contain many menu items and the same menu item can appear in many different orders.

### Part – B

One multivalued attribute can be Customer Phone_No because a customer may have more than one phone number.

One composite attribute can be Restaurant Location because it can be divided into smaller parts such as street, city and state.

**Part – C**

Order_MenuItem(Order_ID, MenuItem_ID, Quantity)

Primary Key:
(Order_ID, MenuItem_ID)

Foreign Keys:
Order_ID → Order(Order_ID)
MenuItem_ID → MenuItem(MenuItem_ID)

Additional Attribute:
Quantity

This table is required because the relationship between Order and MenuItem is many-to-many.

# Part – III

Question – 2

The relationship between Doctor and Patient is many-to-many because a patient may be treated by several doctors during treatment, and a doctor may treat many patients. To represent this relationship, an associative entity called Treatment can be created. The composite primary key of this entity would consist of Doctor_ID and Patient_ID.

In the relationship between Ward and Patient, every patient must be assigned to a ward while admitted to the hospital. Therefore, Patient has total participation in this relationship. A ward may exist even when no patient is currently admitted, so Ward has partial participation.

In the Doctor–Patient relationship, a doctor can exist in the system even if no patients are assigned yet, and a patient can also exist before treatment begins.

Therefore, both Doctor and Patient have partial participation in this relationship.

For attribute design, Patient_Address can be modeled as a composite attribute because it can be divided into House Number, Street, City and State. Storing these parts separately makes searching and managing address information easier and more organized.

Question – 1